

Supplemental Procedures Document

IRBNet Protocol #2446127-1

*Cognitive Warrant: A Framework and Pilot Methodology for
Evaluating Semantic Alignment in Controlled Vocabularies*

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This document is submitted as a supplemental attachment to the IRB application (#2446127-1). It provides detailed procedural information for two components of the study that go beyond the step-by-step summary in Section D.2 of the application form: **Part A** covers the expert consultant validation of LCSH stimulus mappings by two DU librarians, and **Part B** covers pilot testing of the two online surveys with 20 SONA participants. **Part C** provides justification for Exempt Certification. **Part D** provides justification for use of PsychoPy and Pavlovian.

Part A: Expert Consultant Validation of LCSH Stimulus Mappings

Two University of Denver librarians with cataloging experience will serve as expert consultants to validate the principal investigator's mappings of 50 stimulus word senses to authorized Library of Congress Subject Headings (LCSH). The librarians are recruited solely for their professional expertise in subject analysis and authority control. **The librarians are not research participants** as defined under 45 CFR 46.102(e)(1). The study does not investigate the librarians themselves, their behaviors, or their personal characteristics. Rather, their professional judgments are used to evaluate and refine a research instrument and coding procedure. No identifiable private information, biospecimens, demographic data, or personal behavioral data are collected from the librarians. Any agreement statistics (e.g., Cohen's κ) are used exclusively to assess the reliability of the mapping protocol and not to evaluate the librarians as individuals. Accordingly, the librarian validation activity constitutes expert consultation and instrument validation rather than human-subjects research.

A.1. Recruitment

Librarian consultants are identified through a formal request from the principal investigator to the Dean of University of Denver Libraries. The request specifies that librarians with experience in cataloging and, specifically, familiarity with LCSH are sought. The Dean identifies and forwards the request to appropriate library staff; participation is voluntary. The principal investigator has no direct supervisory or evaluative relationship with the librarians and no authority over their employment or professional standing.

A.1.1. Email to Dean of DU Libraries (to be sent *following* IRB approval/exemption)

Dear Dean Levine-Clark,

I am writing to request your support in identifying two librarians from the University of Denver Libraries who would be willing to contribute a small amount of their time to an IRB-approved capstone research project I am completing in the Morgridge College of Education's MLIS program. My advisor, Dr. Krystyna Matusiak, is CC'd on this message and endorses this request.

My project, titled "Cognitive Warrant: A Framework and Pilot Methodology for Evaluating Semantic Alignment in Controlled Vocabularies," investigates whether the categorical and hierarchical structure of LCSH corresponds to measurable patterns in how people cognitively process word meaning. The study uses a stimulus set of 50 word senses — drawn from published psycholinguistic research — each mapped by me to an authorized LCSH heading. Before collecting participant data, I am conducting an inter-rater reliability validation of those mappings, and I need two librarians with cataloging experience to serve as independent expert judges.

What participation would involve: Each librarian would complete two tasks, both of which are straightforward and can be scheduled at their convenience.

Task 1 — Independent Evaluation (approximately 90 minutes to 2 hours, completed individually)

Each librarian would evaluate my mappings of 50 word senses using a Qualtrics interface. For each word sense, they will be shown two short stimulus sentences illustrating a specific use of a word (for example, two sentences illustrating the legal sense of “appeal”) and asked to select the most appropriate LCSH heading from five options. Those options include my selected heading, three other candidate headings I considered during mapping, and an "Other" option with a free-text field for write-in responses. Librarians will also rate their confidence in each selection. The task is self-paced and requires no preparation.

Task 2 — Consensus Meeting (approximately 60 to 90 minutes, held jointly)

Following the independent evaluations, I will calculate inter-rater agreement (Cohen's κ) and compile a list of any items where the two librarians and I diverged. I will then facilitate a brief joint meeting — attended by both librarians and myself — to discuss those disagreements and reach consensus. The length of this meeting depends on the level of initial agreement; I expect it to fall within the 60–90 minute range. This meeting can be held virtually or in person, at the librarians' preference. Total time commitment: approximately 2.5 to 3.5 hours per librarian, scheduled flexibly around their existing responsibilities.

Ideal background: Librarians with experience in original or copy cataloging, subject heading assignment, or familiarity with LCSH would be the strongest fit. Experience with LC's controlled vocabulary is more important than familiarity with the specific research topic.

This study has received exempt certification from the University of Denver Institutional Review Board (IRB Protocol Number 2446127-1), and participating librarians will be acknowledged by name in the final capstone paper, if they consent to acknowledgment.

I would be grateful for any support you can offer in identifying willing participants. I am happy to provide additional information about the study, share the IRB approval documentation, or arrange a brief call if that would be helpful. I anticipate beginning data collection in the coming weeks and would hope to schedule the librarian evaluation within that same timeframe.

Thank you very much for your time and consideration.

Respectfully,

Nicholas Meister

MLIS Candidate, Morgridge College of Education

University of Denver

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cc: Dr. Krystyna Matusiak, Professor, Morgridge College of Education

A.2. Communication and Orientation

Upon identification, each librarian receives an introductory email from the principal investigator that includes:

1. a description of the study and the purpose of the validation task;
2. a complete description of what participation entails, including the independent evaluation task and the follow-up consensus meeting;
3. explicit clarification that they are serving as expert consultants and not as research participants;
4. further clarification that they may withdraw from the consultation process at any time

without penalty;

5. notice that aggregate ratings and qualitative feedback may be described in the methodology section of the capstone paper, but that no personal identifying information will be attributed to them without their explicit consent;
6. a link to the online evaluation interface; and
7. a request that if either librarian is unable to fulfill their role, they inform the Dean of Libraries as soon as practicable so that a replacement can be identified.

A.2.1. Initial Email to Librarians

Dear [Name],

Thank you for agreeing to assist with my capstone research project. I am a student in the MLIS program at the Morgridge College of Education, working under the supervision of Dr. Krystyna Matusiak. I am writing to introduce myself, describe the study and your role in it, and share everything you need to get started.

ABOUT THE STUDY

My capstone project, titled "Cognitive Warrant: A Framework and Pilot Methodology for Evaluating Semantic Alignment in Controlled Vocabularies," investigates whether the categorical and hierarchical structure of Library of Congress Subject Headings corresponds to measurable patterns in how people cognitively process word meaning. In plain terms: I am testing whether the organizational logic built into LCSH — the way headings cluster into categories and relate to one another through broader and narrower terms — reflects the way users actually think about words and concepts.

The study uses 50 word senses drawn from published psycholinguistic research (Trott & Bergen, 2023). Each word sense has been mapped by me to an authorized LCSH heading using a structured decision procedure. Before collecting data from study participants, I need to validate that these mappings are defensible from the perspective of professional cataloging practice. That is where your expertise becomes essential.

YOUR ROLE

You are invited to serve as an expert consultant on this project, not as a research participant. Your role is to independently evaluate my LCSH mappings using your professional judgment, and then to meet with me and the other consulting librarian to discuss any discrepancies. Your cataloging expertise is the point: I am seeking the kind of informed, practice-grounded evaluation that only an experienced cataloger can provide.

This consultation involves two tasks.

Task 1 — Independent Evaluation (approximately 90 minutes to 2 hours, completed on your own schedule)

You will access an evaluation interface on Qualtrics. For each of the 50 word senses, you

will be shown two short sentences that illustrate a specific use of a word — for example, two sentences illustrating the legal sense of the word "appeal" — and asked to select the LCSH heading that best represents the shared sense of the word across both sentences. You will choose from four authorized heading options I have curated, or you may enter a heading of your own if none of the options fit. You will also rate your confidence in each selection and may leave optional notes. The task is self-paced and requires no preparation beforehand.

Task 2 — Consensus Meeting (approximately 60 to 90 minutes, scheduled jointly)

After both consultants have completed the independent evaluation, I will calculate agreement between your responses and mine, and compile a list of any items where our judgments diverged. I will then reach out to schedule a meeting with both of you together. In this meeting, we will review the discrepant items, discuss the reasoning behind each choice, and work toward consensus. The meeting can be held in person or via Zoom, whichever is more convenient. The total time commitment for both tasks combined is approximately 2.5 to 3.5 hours.

A FEW IMPORTANT CLARIFICATIONS

As noted above, you are serving as an expert consultant, not as a research participant. The DU IRB has reviewed this study and approved the consultation process as a standard expert validation procedure.

Your individual responses — heading selections, confidence ratings, and any typed notes — will be used solely by me to calculate inter-rater agreement and refine my mappings. This data will not be published and will be deleted after the project is complete. If the consultation process is described in the methodology section of my capstone paper or any future publication, your involvement will be characterized in the aggregate (e.g., "two DU librarians with cataloging experience") and no personal identifying information will be attributed to you without your explicit prior consent.

You may withdraw from the consultation process at any time without penalty. If, for any reason, you find yourself unable to complete your role in this consultation — whether due to scheduling conflicts, workload, or any other reason — please inform Dean Levine-Clark as soon as practicable so that alternative arrangements can be made in a timely way. I understand that professional obligations shift unexpectedly and there is no need to apologize; I simply ask for as much advance notice as possible.

GETTING STARTED

You can access your evaluation interface at the link below. When you open it, you will find a brief orientation and step-by-step instructions before the first item appears. I recommend completing the evaluation in a single uninterrupted session if your schedule permits, but you may return and revise responses at any time before notifying me that you are finished.

[INSERT QUALTRICS INTERFACE LINK]

Please reply to this email to confirm receipt. When you have completed the evaluation,

email me a brief note so I can calculate agreement and schedule the consensus meeting.

I am grateful for your time and expertise, and I welcome any questions you may have about the study or the task. Please do not hesitate to reach out.

Respectfully,

Nicholas Meister

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University of Denver

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Faculty Advisor: Dr. Krystyna Matusiak | krystyna.matusiak@du.edu

A.3. Task 1 — Independent Evaluation

Duration: approximately 90 minutes to 2 hours per librarian; completed asynchronously.

Each librarian independently completes an online evaluation using an interface hosted in Qualtrics. The two librarians complete this task separately and are not aware of each other's responses during this phase.

For each of the 50 stimulus word senses in the study's stratified sample, the librarian is presented with:

- two short English sentences that together illustrate a specific sense of an ambiguous word (**note:** all sentences that the librarians will see are included as a separate file in the IRB application);
- three to four authorized LCSH heading options (labeled A–D), drawn from the principal investigator's candidate heading pool for that word sense, including the PI's final mapping selection and two to three alternative candidates; option letter assignments are randomized across items to prevent position bias; and
- a final option (“Other”) with a free-text field, allowing the librarian to enter a different authorized heading if none of the previous options is satisfactory in their professional judgment.

As illustrated in Figure 1, for each item the librarian: (a) selects the heading that best represents the shared sense of the word as established by the two sentences; (b) rates their confidence in that selection on a 1–5 scale; and (c) optionally enters a brief written justification or note. Librarians mark each item complete before advancing to the next. The task is self-paced and may be completed in one session or across multiple sessions within the agreed evaluation window.

The interface does not collect any demographic information, browsing data, or personal identifying information beyond the librarian's randomized ID as pre-assigned by the principal

Word Meanings and Descriptive Labels Study

Progress:
14 of 50

appeal

POLYSEMY

Item 14 of 50

STIMULUS SENTENCES

S1 He had a legal **appeal**.

S2 He had a pending **appeal**.

HEADING OPTIONS

Select the heading that best represents the sense of “appeal” as used in the sentences above.

☒ **A**

Appellate procedure

id.loc.gov/authorities/subjects/sh85006085

Variant: Appeal. **Broader Terms:** Courts; Procedure (Law). **Narrower Terms:** Appeal as from an abuse; Appeals in forma pauperis; Certiorari; Courts of last resort; Courts of last resort; Exceptions (Law); Law and fact; New matter (Civil procedure); New trials; Prospective overruling; Reformatio in pejus; Tax protests and appeals; Waiver of appeal; Writ of error. **Related Terms:** Appellate courts; Civil procedure; Criminal procedure; Trial practice

☐ **B**

Appellate courts

id.loc.gov/authorities/subjects/sh85101308

Variants: Courts of appeals; Supreme Courts. **Broader Term:** Courts. **Related Terms:** Appellate procedure; Courts of last resort

☐ **C**

Court proceedings

id.loc.gov/authorities/subjects/sh2018003244

Variants: Courtroom proceedings; Proceedings, Court. **Broader Term:** Justice, Administration of. **Narrower Terms:** Actions and defenses; Audiotapes in court proceedings; Closed-circuit television in court proceedings; Complex litigation; Conduct of court proceedings; Government litigation; Television broadcasting of court proceedings; Trials; Video tapes in court proceedings.

☐ **D**

Petition, Right of

id.loc.gov/authorities/subjects/sh85100334

Scope notes: Here are entered works on the right of citizens to petition the government for redress of grievances. Works on the English procedure for claims against the crown are entered under [Petition of right.] **Variant:** Right of petition. **Broader Term:** Political rights

☐ **E**

Other — enter authorized heading below

If none of the options above is appropriate, enter the authorized LCSH heading as it appears in the LC authority file.

Enter authorized heading here...

YOUR RESPONSE

Confidence in your selection — required

1

2

3

4

5

Very uncertain

Very confident

Notes — optional, but encouraged

e.g., “Option B is also plausible given the administrative context.” or “The sentences are ambiguous between the legal and the procedural sense.” Observations about close calls, borderline fits, or concerns about the heading options are particularly useful.

← Previous

☐ Mark item as complete before advancing

Next item →

Figure 1: Mock-up of the Qualtrics-based librarian evaluation interface. Each item presents the target word, its ambiguity type (here: polysemy), and two stimulus sentences establishing the specific word sense under evaluation. The consultant selects the best-fitting authorized LCSH heading from four labeled options (A–D), each displayed with its URI and key information from its LC authority file, or enters an alternative heading via option E. A required confidence rating (1–5) and optional free-text notes field are present. A progress bar tracks completion across all 50 items. Shown here: *appeal*, legal/procedural sense (item 14 of 50); option A (*Appellate procedure*) is selected with a confidence rating of 3.

investigator when configuring their individual evaluation queue.

A.3.1. Evaluation criteria

To support consistency across evaluations, librarians are oriented to the same criteria used by the PI during the original mapping process. These are provided for reference, not as a formal scoring procedure.

Criterion	Question to ask
Semantic fit	Does this heading match the specific sense established by the two sentences?
Scope note alignment	If a scope note is present, does this use of the word fall within it?
Hierarchical coherence	Are the broader and narrower terms of this heading consistent with the conceptual domain of the word sense?
Specificity	Is this heading appropriately specific — neither too broad nor too narrow for the sense as illustrated?
UF coverage	Do the heading’s variant terms (<i>Use For</i> entries) suggest it is intended to capture this type of usage?

The most important criterion is *semantic fit*: the heading should represent the domain of the word sense as the sentences establish it, not the word itself in the abstract.

A.3.2. Data recorded per item

Field	What is recorded
Selection (A–E)	The letter of the chosen heading, or “Other”
Write-in heading	Authorized LCSH string, if “Other” is selected
Confidence (1–5)	1 = very uncertain; 3 = moderately confident; 5 = very confident
Optional notes	Observations about the item — e.g., competing headings, ambiguous sentences, scope note concerns
Completion status	Item marked complete before advancing

A.4. Inter-Rater Agreement Calculation

Following completion of both independent evaluations, the principal investigator calculates Cohen’s kappa (Cohen, 1960) as the primary measure of categorical inter-rater agreement. Percent agreement is also reported alongside kappa. A threshold of $\kappa \geq 0.80$ is required

before proceeding with full stimulus mapping (Landis & Koch, 1977; Hallgren, 2012). Items where the two librarians diverge from each other or from the principal investigator’s mapping are compiled into a structured disagreement log for use in the consensus meeting.

A.5. Task 2 — Consensus Meeting

Duration: approximately 60 to 90 minutes; conducted in person at the Anderson Academic Commons or via Zoom at the librarians’ preference.

Following the independent evaluations and agreement calculation, the principal investigator schedules a joint meeting with both librarians to review discrepancies and reach consensus on disputed items. The structured disagreement log is shared with both consultants in advance of the meeting so that they may review disputed items privately before convening.

A.5.1. Meeting procedure

For each disputed item, the principal investigator presents the word sense and the nature of the discrepancy without characterizing either consultant’s response as correct or preferred. Each consultant is invited to explain the reasoning behind their selection. The group then discusses the structural and semantic features of the competing headings — drawing on LCSH scope notes, broader and narrower term relationships, and the mapping rubric used by the principal investigator — until consensus is reached. The principal investigator documents the consensus heading and a brief rationale for each resolved item. No consultant will be pressured to change their professional judgment; if consensus cannot be reached on a given item, that item will be flagged for exclusion from the analysis.

At the close of the meeting, both consultants are invited to share general observations about the heading options, the mapping criteria, or the task design. This qualitative feedback is voluntary and will be used solely to refine the mapping guidelines for the full stimulus set.

A.5.2. Recording policy

No recording of the meeting will be made without the explicit consent of both consultants. If neither consultant consents to recording, the principal investigator will only take written notes of consensus headings, rationale for each, and qualitative feedback.

A.5.3. Follow-up

Following the meeting, the principal investigator may send brief clarifying questions to individual librarians via email. No consultant is obligated to respond.

A.6. Duration and Scheduling

The total time commitment for each librarian consultant is approximately 2.5 to 3.5 hours: approximately 90 minutes to 2 hours for the independent evaluation (Task 1) and approximately 60 to 90 minutes for the consensus meeting (Task 2). Task 1 is completed asynchronously at the librarian's convenience within an agreed window. Task 2 is scheduled mutually at a time that does not conflict with the librarians' regular responsibilities.

A.7. Supplemental Responses: D.2 and H.2 (Librarian Consultants)

A.7.1. D.2 — Procedural summary for librarian consultation

The following supplements the step-by-step description in Section D.2 of the application form.

Phase 2 — Expert Consultant Validation of LCSH Stimulus Mappings (approx. 2–3 weeks). Following proof-of-concept analysis of existing data, the PI completes LCSH mappings for all 50 word senses in the study's stratified sample. Two University of Denver librarians with cataloging experience are engaged as expert consultants through a formal request to the Dean of University of Denver Libraries. Each librarian independently accesses a purpose-built Qualtrics interface and evaluates all 50 mappings per Section A.3 above (Task 1, asynchronous). The PI then calculates inter-rater agreement. A structured consensus meeting is conducted to resolve discrepancies (Task 2; Anderson Academic Commons or Zoom). Conducted by: Nicholas Meister (PI), supervised by Dr. Krystyna Matusiak.

A.7.2. H.2 — Risks to librarian consultants

Non-human subjects determination. The librarian validation activity constitutes expert consultation and instrument validation rather than human-subjects research under 45 CFR 46.102(e). The following risk statement is provided as a matter of professional ethics and research integrity.

Anticipated risks. Participation as an expert consultant presents minimal risk. Librarians are asked to exercise professional judgment on a domain-specific task consistent with their cataloging expertise. No sensitive, personal, or clinical information is collected from them. No deception is used. The evaluation task involves no physical procedures, no psychological stress, and no material that could be considered sensitive, distressing, or controversial. Participation is voluntary; any librarian may withdraw at any time without consequence to their employment or professional standing.

Confidentiality protections.

- *Response data.* Individual heading selections, confidence ratings, and typed justifications are stored in a password-protected Qualtrics account accessible only to the principal investigator. Raw response data are not shared with other study personnel, will not be

published, and will be deleted after the capstone project is complete.

- *Written products.* If librarian involvement is described in the methodology section of the capstone paper or any subsequent publication, it will be characterized in the aggregate (e.g., “two DU librarians with cataloging experience”) rather than by name, unless a librarian provides explicit written consent to individual acknowledgment.
- *Consensus meeting notes.* Meeting notes record consensus decisions and their rationale but do not attribute specific statements to individual consultants. Notes are not published and will be deleted after the project is complete.
- *Follow-up correspondence.* Email correspondence with librarians is retained only as long as necessary to resolve outstanding questions and is not published or included in study data.
- *Data retention.* All librarian-associated data (response records, meeting notes, email correspondence) are stored on the principal investigator’s password-protected university-affiliated accounts and will be destroyed no later than one year following completion of the capstone project.

References cited in this section:

- Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1), 37–46.
- Hallgren, K. A. (2012). Computing inter-rater reliability for observational data. *Tutorials in Quantitative Methods for Psychology*, 8(1), 23–34.
- Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–174.

Part B: Pilot Testing of Online Surveys

Overview. Pilot testing will be conducted of two online surveys with **20 participants** (minimum reporting threshold: 15). Refinements will be made based on qualitative feedback through a brief post-task survey administered at the end of the session. *Where:* Online via the Pavlovia platform (participants complete surveys remotely on their own computers). *By whom:* Nicholas Meister (PI), supervised by Dr. Krystyna Matusiak. Survey 1 replicates Trott and Bergen’s (2023) sensibility judgment paradigm, substituting LCSH categorical boundaries for dictionary sense boundaries. Survey 2 introduces an appropriateness rating paradigm that directly measures cataloger–user alignment. Session duration: approximately 45–55 minutes total; see Section B.2 for a complete trial structure and session breakdown.

B.1. Pilot Test Goals

The pilot test is designed to collect diagnostic data prior to main data collection (deferred to a subsequent IRB application). Specific goals are:

- **Identify and resolve technical issues.** Verify that Pavlovia task files load correctly across common browsers (Chrome, Firefox), that response recording functions as expected in both surveys, and that the Pavlovia data pipeline produces well-formed output files.
- **Verify stimulus presentation and trial logic.** Confirm that sentence pairs, subject headings, foil conditions, and fixation timing display as intended, and that counterbalancing and stratified stimulus assignment execute correctly across versions.
- **Obtain a realistic session time estimate.** Record total and per-survey session duration to verify against the 45–55 minute estimate stated in the consent form and SONA posting, and to calibrate the break screen placement.
- **Obtain anonymized qualitative feedback.** Collect open-ended responses regarding instruction intelligibility, response format naturalness, and points of confusion — particularly regarding the appropriateness rating scale anchors and the simultaneous sentence presentation format in survey 2.
- **Verify attention check performance.** Confirm that catch trials with clearly inappropriate headings produce the expected response pattern and assess whether the attention check threshold is appropriately calibrated.
- **Review preliminary response distributions.** Inspect response time distributions from survey 1 for outliers, floor effects, and anticipatory responses; inspect survey 2 rating distributions for ceiling/floor effects, restricted range, or bimodality indicating

stimulus calibration problems.

- **Confirm data file structure and completeness.** Verify that Pavlovia output files contain all expected columns (including `ambiguity_type`, `same_heading`, `distance_bert`, and randomized subject ID fields) and that no trial data are silently dropped during the session.

B.2. Study Design, Participant Enrollment, and Trial Structure

B.2.1. Sample Size and Rationale

The pilot test will enroll **20 participants** recruited through the DU Psychology Department’s Research Participation System (SONA), with a **minimum reporting threshold of 15 participants**. Four considerations motivate this enrollment target.

Counterbalancing integrity. Each participant completes both surveys, and survey presentation order is counterbalanced across participants (see Section B.2.6). Twenty participants afford a clean 10/10 split between the two presentation orders. Smaller samples produce unbalanced groups that complicate interpretation of any order-related differences observed in the pilot data.

Observation density for effect size estimation. The two behavioral pilot hypotheses—a continuous semantic distance effect and an LCSH categorical boundary effect—are evaluated descriptively from trial-level data. At 50 critical trials per participant, 20 participants yield 1,000 observations per survey. This is the practical floor for obtaining effect size point estimates stable enough to serve as inputs to a subsequent statistical power analysis for the full study via `simr` in R. Below approximately 750 observations, effect size estimates carry confidence intervals spanning multiple conventional benchmarks (e.g., Cohen’s d spanning the small-to-large range), degrading their utility for power planning. Inferential conclusions are not drawn from pilot behavioral data at any sample size; all behavioral analyses are reported descriptively as feasibility indicators.

Methodological precedent. A pilot of 20 participants is consistent with established norms for paradigm-validation pilots in behavioral psycholinguistics. Trott and Bergen (2023), whose Experiment 1 paradigm Survey 1 directly replicates, employed a pilot of 20 participants before scaling to their full study sample.

Attrition buffer. Recruiting 20 participants provides a buffer against exclusions due to attention check failure, incomplete sessions, or technical errors on the Pavlovia platform. If up to five participants are excluded, the dataset remains at or above the minimum reporting threshold of 15.

B.2.2. Full Stimulus Overlap Across Surveys

Every participant completes both surveys using the complete set of 50 critical stimuli. An alternative split design—in which each participant sees a different 25 stimuli in each survey—was considered and rejected. First, a split design would halve observations per stimulus per survey without reducing session length, yielding approximately 375 observations per survey at $N = 15$ rather than 750. Second, a split design structurally prevents any item-level comparison of Survey 1 and Survey 2 responses for the same stimulus, eliminating a diagnostic check that the full-overlap design preserves at no additional cost. Third, split-design assignment would require managing stimulus counterbalancing across participants—an implementation burden disproportionate to the pilot’s diagnostic goals.

Carryover effects between surveys are mitigated by counterbalancing presentation order (Section B.2.6) and by the cognitive dissimilarity of the two tasks: Survey 1 involves rapid, implicit keypress responses to sentence pairs, while Survey 2 involves deliberate, explicit rating of heading appropriateness. Participants are unlikely to form retrievable episodic memories from Survey 1’s rapid sensibility-judgment sequence that would meaningfully bias Survey 2 ratings. If order-related differences are observed in the pilot data, they will be reported descriptively as a design consideration for the full study.

B.2.3. Survey 1 — Primed Sensibility Judgment

Critical trials: 50. The pilot’s critical stimulus set comprises 25 ambiguous English words, each with two senses and two stimulus sentences per sense (100 sentences total; 150 possible sentence pairs). For each word, two sentence pairs are selected for presentation: (1) one *same-sense* pair, in which both the prime and target sentence are drawn from the same word sense (`same_heading` = 1 by definition); and (2) one *different-sense* pair, in which the prime and target are drawn from different senses (`same_heading` = 0 for the majority of pairs, except collapse cases in which both senses map to the same LCSH authority heading). This design yields 25 trials in each level of `same_heading` per participant—the maximally balanced configuration for evaluating the categorical boundary hypothesis. For each participant, the specific cross-sense pairing used is selected via runtime randomization from the four possible combinations, so that the full range of cross-sense pairings is represented across the dataset.

Filler trials: 50. One sentence pair is presented per filler word ($N = 50$ filler words). Approximately half of filler target sentences are nonsensical (i.e., the correct response is “does not make sense”). This proportion keeps the task calibrated without inflating the base rate of “no” responses to a level that might alert participants to the critical manipulation. The 50:50 critical-to-filler ratio matches Trott and Bergen’s (2023) design.

Total: 100 trials. A single mid-task break screen is presented after trial 50. Estimated duration: 8–10 minutes.

B.2.4. Survey 2 — Heading Appropriateness Rating

Critical trials: 50. One trial is presented per word sense (50 senses total). Each trial presents two sense-establishing sentences simultaneously and asks the participant to rate four LCSH heading options on a 1–7 appropriateness scale. The four options for each trial are:

- (1) the *target heading* (the PI’s LCSH mapping for that sense);
- (2) a *within-neighborhood foil* (`same_nbhd` = 1; a heading from the same hierarchical neighborhood as the target, sharing a common broader-term ancestor within three levels);
- (3) a *cross-neighborhood foil* (`same_nbhd` = 0; a heading from a different hierarchical section of LCSH); and
- (4) a *clearly inappropriate heading* serving as an embedded attention check anchor.

Participants who rate the clearly inappropriate heading at 4 or higher on more than 30% of critical trials are flagged for exclusion.

Filler and catch trials: 10. Ten additional trials use filler word senses (drawn from the pilot’s 50-item filler bank) for which the appropriate LCSH heading is unambiguous. These reinforce task engagement and provide a secondary attention check independent of the embedded anchors in critical trials.

Total: 60 trials. Estimated duration: 25–28 minutes.

B.2.5. Session Summary

Component	Trials	Est. Duration
Eligibility screening, consent	—	~4 min
Survey A (first, per counterbalance group)	100 <i>or</i> 60	~10 <i>or</i> 28 min
Between-survey break and transition screen	—	~2 min
Survey B (second, per counterbalance group)	60 <i>or</i> 100	~28 <i>or</i> 10 min
Demographics	—	~1 min
Post-task qualitative feedback survey	—	~4 min
Total		~45–55 min

The consent form and SONA study posting state a session duration of 45–55 minutes. The wider window reflects genuine variation in Survey 2 completion time, which depends on self-paced reading of simultaneous sentence pairs and deliberate heading evaluation.

B.2.6. Experiment Order Counterbalancing

Survey presentation order is counterbalanced across participants. Participants are assigned to groups by alternating SONA sign-up slot: odd-numbered sign-ups are assigned to

Group A, even-numbered to Group B, producing a 10/10 split without manual assignment.

Group	Presentation Order	<i>N</i>
A	Survey 1 → Survey 2	10
B	Survey 2 → Survey 1	10

The Survey 2 → Survey 1 order (Group B) carries a higher theoretical carryover risk: having explicitly evaluated LCSH headings for each word sense in Survey 2 could pre-activate categorical information and influence Survey 1 reaction times. The Survey 1 → Survey 2 order (Group A) is lower risk, as implicit sensibility judgments are unlikely to generate conscious categorical associations transferable to the rating task. Equal group sizes ensure that any carryover effects are balanced across the dataset rather than confounded with condition effects. Asymmetric results across groups will be reported descriptively and treated as a design consideration for the full study. The pilot is not powered to test an Order × Condition interaction inferentially.

B.3. SONA Recruitment and Eligibility Screening

B.3.1. Recruitment

Potential study participants will find the following information on the SONA site.

Field	Entry
Study Name	Word Meanings and Descriptive Labels Study
Study Type	This is an online study on another website. To participate, sign up, and then you will be given access to the website to participate in the study.
Duration	45–55 minutes total
Abstract	You will read short sentences and judge whether they make grammatical sense and also rate how well category labels describe the meaning of words used in those sentences.

(continued on next page)

(continued from previous page)

Field	Entry
Description	The purpose of this study is to investigate how people understand words that have more than one meaning. The study is part of research on language comprehension and the cognitive organization of word meaning. In two online surveys, you will read a series of short sentences containing common English words. In one portion, your task is to indicate which sentences make sense grammatically and which do not by pressing the appropriate key. In another portion of the study, you will evaluate how well certain descriptive labels fit the content of sentence pairs and select an appropriateness rating on a scale of 1–7. If you agree to participate after reading the consent form, you will be sent a link to the surveys on the Pavlovia platform. You have the option to either start right away or return to the study at a later time. The study will take 45-55 minutes and should be completed in a quiet and private space. Study completion must be completed in a single setting in order to receive Chits. Please do not proceed if you are not able to complete the study in a single sitting at this time. After you have completed your study participation, 2 Chits will be assigned to your SONA account within 2 business days.
Eligibility Requirements	(1) 18 years of age or older; (2) native English speaker or have native-level English proficiency; (3) able to complete the study on a computer or laptop (not a phone or tablet) with a keyboard and internet connection; (4) able to read English text comfortably without significant visual or reading difficulties.
Website	You may not view the website until you sign up for this study.
Researchers	Nicholas Meister, Krystyna Matusiak
Principal Investigator	Nicholas Meister

B.3.2. Eligibility Screening

Potential participants encounter the following eligibility screening instrument, administered through SONA prior to the informed consent form.

Purpose of the screening instrument. This screening instrument is administered before the informed consent form to determine whether prospective participants meet the eligibility criteria for this study. Participants must answer all questions. Responses are used solely for eligibility assessment and are not included in any study data.

You will be notified if you are eligible for this study. The information you provide below will **not** be used in the study itself. If you are not eligible for this study, you will be notified, and any information you provide below will not be retained.

B.3.3. Inclusion criteria — all four must be answered “Yes” to proceed

Question	Text
Q1 of 4	Are you 18 years of age or older?
Q2 of 4	Are you a native English speaker, or do you have native-level English proficiency?
Q3 of 4	Are you completing this study on a computer or laptop (not a phone or tablet) with a keyboard?
Q4 of 4	Can you read English text comfortably without significant visual or reading difficulties?

B.3.4. Eligibility determination

To be eligible, a participant must answer “Yes” to Questions 1–4. Ineligible participants are informed they do not meet the study criteria. No personally identifiable information is retained for ineligible participants, and all screening data from ineligible individuals are destroyed. Disqualifying responses trigger automatic session termination; eligible participants are automatically advanced to the informed consent form. Participant IDs are anonymized, and SONA identities are not linked to experimental data.

B.4. Step-by-Step Procedures: What Participants Encounter

Pilot testing is conducted entirely online via the Pavlovia platform. Participants complete all procedures remotely on their own computers. All research is conducted by Nicholas Meister (PI), supervised by Dr. Krystyna Matusiak, with no in-person contact between researchers and participants. Survey presentation order is counterbalanced between participants; see Section B.2.6. Participants may take a brief, self-paced break between surveys.

B.4.1. Step 1 — Eligibility screening (approx. 2 minutes; administered via SONA)

Described in Section B.3 above.

B.4.2. Step 2 — Informed consent (approx. 2 minutes)

Eligible participants review and complete the online survey consent form before proceeding to any survey task. The form is included in the IRB application as a separate file.

B.4.3. Step 3 — Survey 1: Sensibility Judgment Task (approx. 8–10 minutes)

Participants complete a computerized sensibility judgment task hosted on Pavlovia. Before the main task, participants read the task instructions and complete four practice trials using sentences unrelated to the study stimuli.

Participant instructions.

In this session, you will read pairs of short sentences presented one at a time on the screen. Your task is to decide whether each sentence makes sense. Read the sentence carefully. Decide whether the sentence is sensible or not (grammatically). Please respond quickly without sacrificing accuracy. Press **F** for YES (the sentence makes sense). Press **J** for NO (the sentence does not make sense). Some sentence pairs may seem very similar, while others may involve different meanings of the same word. This session will take approximately 8–10 minutes.

Practice items.

Sentence	Expected Response
He saw the big dog.	Yes (sensible)
He saw the sky dog.	No (nonsensical)
It was a square circle.	No (nonsensical)
It was a large circle.	Yes (sensible)

Trial flow. Each trial follows this fixed sequence (see Figure 2):

1. Fixation cross: 500 ms
2. Prime sentence displayed until response (maximum 15 s)
3. Participant presses **F** or **J**
4. Fixation cross: 500 ms
5. Target sentence displayed until response (maximum 15 s)
6. Participant presses **F** or **J**
7. Inter-trial interval: 1,000 ms

Trials are presented in randomized order. The session includes both sensible and nonsensical sentences. Primary measures are response time (milliseconds) and response accuracy.

Data collected (Survey 1). Participant ID (randomized); trial number; stimulus word; prime sentence condition; target sentence condition; response key pressed; accuracy; response time (ms); survey completion time.

B.4.4. Step 4 — Between-survey break and transition screen (approx. 2 minutes)

B.4.5. Step 5 — Survey 2: LCSH Appropriateness Rating Task (approx. 25–28 minutes)

Participants complete a second computerized task hosted on Pavlovia. Participants first complete 3 practice items to become familiar with the rating procedure.

Participant instructions.

In this session, you will read pairs of short sentences containing the same word. After each sentence pair, you will see several proposed library subject headings. Your task is to rate how well each subject heading matches the meaning of the word in the sentences. Rate each subject heading independently (i.e., rate each heading on its own merits, not relative to the others). Please use the following scale: 1 = Very poor match; 2 = Poor match; 3 = Slightly poor match; 4 = Neutral / Unsure; 5 = Slightly good match; 6 = Good match; 7 = Very good match. There are no right or wrong answers. Please respond based on your own judgment. This session will take approximately 25–28 minutes.

Practice items.

Sentence	Proposed heading	Expected Pattern
He saw a furry bat.	Bats	High
He saw a furry bat.	Baseball bats	Low
She filed a legal appeal.	Appellate procedure	High
She filed a legal appeal.	Interpersonal attraction	Low

Subject heading alternatives — generation protocol. Each stimulus sentence pair is presented with three or four subject headings serving distinct methodological roles. Only heading types 1–3 enter the Δ_r (discrimination score) computation. The attention checker (type 4) is reserved for data quality screening.

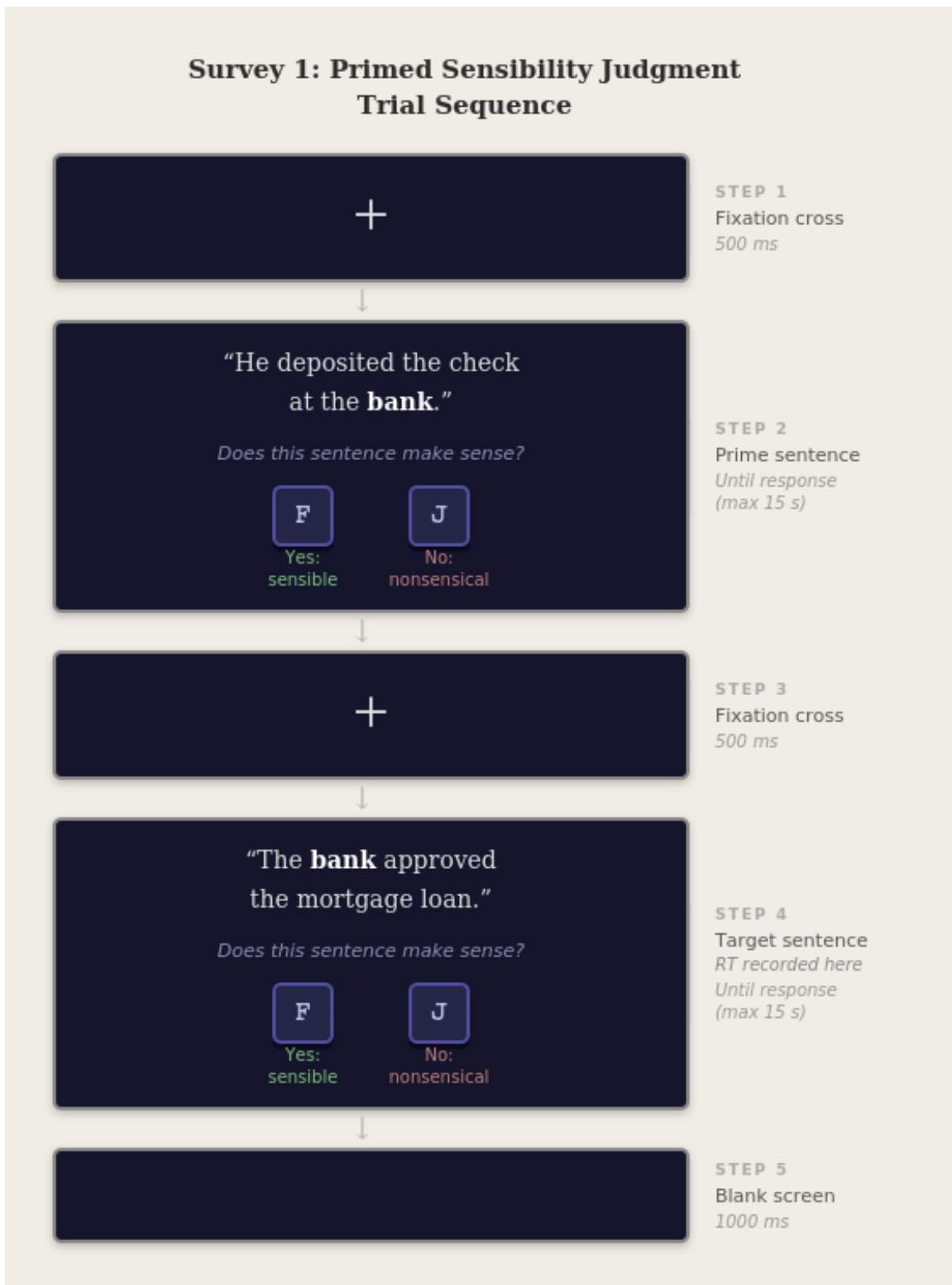


Figure 2: Mock-up of the interface participants will encounter in survey 1.

Type	Label	Description
1	Target heading	Matches the word sense instantiated by the sentence pair, validated through librarian consultation. BERT cosine distance from sentence-pair embedding: 0.0–0.2.
2	Within-category foil	Drawn from the same superordinate LCSH category as the target (shares a Broader Term); inappropriate for the specific sense but topically coherent. BERT cosine distance: 0.4–0.7, matched to type 3.
3	Cross-category foil	Drawn from a different LCSH category; for polysemous targets, corresponds to an <i>alternative</i> sense of the word. BERT cosine distance: 0.4–0.7, matched to type 2.
4	Attention checker	Completely unrelated to any plausible interpretation of the stimulus. BERT cosine distance: 0.8–1.0. Used for exclusion criterion only; excluded from all primary analyses.

BERT cosine distance: A quantitative measure of semantic distance (i.e., dissimilarity) between word senses, operationalized as the cosine distance between BERT-derived contextualized word embeddings. Formally, cosine distance = $1 - \text{cosine similarity}$, where cosine similarity $\in [-1, 1]$, yielding a cosine distance range of $[0, 2]$: proportional vectors have distance 0, orthogonal vectors have distance 1, and opposite vectors have distance 2. BERT (Bidirectional Encoder Representations from Transformers) is a large language model that generates word embeddings capturing meaning-in-context rather than static word meanings; the resulting contextualized vectors enable measurement of continuous semantic dissimilarity independently of categorical sense assignments, and constitute the continuous predictor (`distance_bert`) throughout this study.

Trial flow. Simultaneous (same-screen rating) design (see Figure 3). The sentence pair is shown once; the participant rates all headings (target + 1–2 foils + attention checker) on the same screen. All headings are presented simultaneously; heading position is randomized. Participants are instructed to rate each heading independently.

1. Two stimulus sentences displayed at top of screen
2. Proposed LCSH headings displayed below, each with a 7-point rating scale
3. Participant clicks a number from 1–7 for each heading
4. Participant clicks Submit to advance to next item

The simultaneous collection of ratings will be reported as a limitation of the pilot study design.

Counterbalancing. Between subjects: presentation order of context sentences is randomized; order of headings within each context is randomized; which heading appears first/last is balanced. *Within subjects:* each participant sees a mix of all subject heading types.

Data collected (Survey 2). Participant ID (randomized); item number; stimulus sentences; ambiguous word; proposed LCSH headings; ratings selected (1–7); response time (optional); completion time.

B.4.6. Step 6 — Demographics questionnaire (approx. 1 minute)

At the end of the surveys, participants will complete a brief demographic questionnaire that includes:

1. age
2. gender
3. Native English speaker or native-level English proficiency (Yes/No)
4. familiarity with library subject headings

Identifying participants with cataloging training or experience instead of excluding them

The specialized knowledge of professional catalogers or individuals with extensive training in LCSH could confound results (though general familiarity with library subject headings is acceptable). We first considered excluding these individuals, but the case for removing the exclusion criterion is strong. The SONA pool is undergraduate psychology students, and the base rate of anyone with professional cataloging experience or formal LCSH coursework in that population is effectively zero. Maintaining the exclusion criterion in the recruitment materials costs us more in hypothesis priming than it saves us in data quality — we are excluding a phantom threat while advertising the research goal to everyone who reads the posting.

The post-hoc flagging approach in the demographic questionnaire is the standard solution in psycholinguistics. Collecting domain familiarity as a covariate and flagging high scorers for sensitivity analysis rather than excluding them a priori is methodologically cleaner. It preserves our full sample, gives us data to report, and lets us demonstrate empirically that the result holds with or without those participants rather than simply asserting it would.

For Item 4, however, a 7-point scale anchored at "no familiarity" and "extensive professional experience" may cluster almost all responses at 1–2 given the population, giving us limited variance to work with. We will get more useful discrimination with a forced-choice descriptor list:

Which of the following best describes your familiarity with library subject headings? (Select

LCSH Heading Appropriateness Rating

Rate how well each LCSH subject heading would describe a document on the topic of the sentences below.

Rate each heading independently on its own merits.

CONTEXT

“She walked into the **bank** to deposit her paycheck.”

“The **bank** approved his application for a small business loan.”

<i>LCSH Heading</i>	<i>Not appropriate</i>	1	2	3	4	5	6	<i>Perfectly appropriate</i>
<i>Banks and banking</i>		☐	☐	☐	☐	☐	☐	☐
<i>Credit unions</i>		☐	☐	☐	☐	☐	☐	☐
<i>Riparian areas</i>		☐	☐	☐	☐	☐	☐	☐
<i>Butterfly gardening</i>		☐	☐	☐	☐	☐	☐	☐

Submit

Figure 3: Mock-up of the interface participants will encounter in survey 2.

one description)

1. I have never heard of library subject headings
2. I have a general awareness of them but no formal training
3. I have used them in a library or research context
4. I have taken coursework that covered library subject headings
5. I have professional cataloging experience

This gives us an ordinal variable with meaningful categories, flags the rare high-familiarity participant unambiguously, and avoids the floor-effect problem of a 7-point scale on a near-zero-variance population.

B.4.7. Step 7 — Qualitative feedback survey (approx. 4 minutes)

After the demographic questionnaire, participants complete a brief anonymous qualitative feedback survey administered within the Pavlovian session. The survey collects open-ended and scaled responses about technical performance, session length, instruction clarity, response format naturalness, and any sources of confusion encountered (see section B.5 below). Participants are informed prior to the survey that responses are anonymous and will not affect their SONA credit. Responses are used solely for methodological refinement prior to the main study.

B.4.8. Step 8 — Debriefing

Participants receive the following debriefing statement. The PI's contact information is provided for any follow-up questions.

Thank you for participating in this study.

Now that you have completed both tasks, we can share the full purpose of the research. This study investigates how people understand words that have more than one meaning — and whether a widely used library classification system reflects the way the human mind organizes those meanings.

In one task, you judged whether sentences made sense. We were measuring how quickly you responded, because response time is an indicator of how easily the brain connects two related uses of the same word. In the other task, you rated how well certain subject headings described the content of sentence pairs. Those headings come from the Library of Congress Subject Headings (LCSH), the system that librarians use to classify and organize library collections. We are testing whether the categories that librarians have assigned to different word meanings match the categories that emerge naturally from human language processing — in other words, whether the library's organization of meaning is one that the mind finds intuitive.

You were not told the full purpose of the study beforehand because prior knowl-

edge of what we were measuring might have affected your responses. Your data are still fully usable and are genuinely valuable to the research.

If you have any questions about this study, please contact Nicholas Meister at nicholas.meister@du.edu.

Total session duration: approximately 45–55 minutes.

B.5. Qualitative Feedback Survey Questions

The qualitative survey is introduced with a screen explaining that responses are anonymous, will not affect SONA credit, and that candid feedback is the goal. Each participant is presented with a subset of the following questions (the numbering of surveys will depend on the order they were presented to each participant).

B.5.1. General / Technical

GT-1.

Did you experience any technical problems during the session? (*No problems / Minor problems that did not affect my responses / Problems that may have affected my responses / I am unsure*)

GT-2.

If you selected anything other than “No problems,” please briefly describe what occurred. (*Open text; conditional on GT-1*)

GT-3.

Which browser were you using? (*Chrome / Firefox / Safari / Edge / Other*)

GT-4.

How would you describe the overall length of the session? (*5-point: Much too short — About right — Much too long*)

B.5.2. Survey 1: Sensibility Judgment Task

E1-1.

How clear were the instructions for the [first/second] task? (*5-point: Very unclear — Very clear*)

E1-2.

Did anything about the instructions for this task confuse you or require re-reading? (*Yes / No / Unsure*)

E1-3.

If yes, please describe what was confusing. (*Open text; conditional on E1-2*)

E1-4.

How natural did the keyboard response format feel (pressing F/J)? (*5-point: Very*

unnatural — Very natural)

E1-5.

Did the pacing of the trials feel appropriate? (*5-point: Much too fast — About right — Much too slow*)

E1-6.

Were there sentences you found genuinely difficult to judge as sensible or nonsensical? (*Yes / No / Unsure*)

E1-7.

If yes, please give an example or describe what made it difficult. (*Open text; conditional on E1-6*)

B.5.3. Survey 2: Subject Heading Appropriateness Rating Task

E2-1.

How clear were the instructions for the [first/second] task? (*5-point: Very unclear — Very clear*)

E2-2.

Did anything about the instructions confuse you or require re-reading? (*Yes / No / Unsure*)

E2-3.

If yes, please describe what was confusing. (*Open text; conditional on E2-2*)

E2-4.

How easy was the 1–7 rating scale to use? (*5-point: Very difficult — Very easy*)

E2-5.

Were the meanings of the scale anchors (1 = “very poor match,” 7 = “very good match”) clear throughout the task? (*5-point: Not at all clear — Perfectly clear*)

E2-6.

Were the subject headings generally understandable as phrases, even if unfamiliar? (*5-point: Almost none — Almost all*)

E2-7.

Did you encounter any subject headings that were confusing, ambiguous, or difficult to evaluate? (*Yes / No / Unsure*)

E2-8.

If yes, describe the heading or source of confusion as best you can recall. (*Open text; conditional on E2-7*)

E2-9.

Was the presentation of the sentence and subject heading together on the same screen clear? (*5-point: Not at all clear — Perfectly clear*)

E2-10.

How would you describe the pacing of this task? (*5-point: Much too fast — About right — Much too slow*)

B.5.4. General Feedback**GF-1.**

Were there any moments during the session where you were unsure what you were supposed to do? (*Yes / No*)

GF-2.

If yes, please describe when and what caused the uncertainty. (*Open text; conditional on GF-1*)

GF-3.

Is there anything about the task instructions, response format, or session structure that you would change? (*Open text*)

GF-4.

Is there anything else you would like the researcher to know about your experience? (*Open text*)

B.6. Supplemental Responses: D.2 and H.2 (Pilot Participants)**B.6.1. D.2 — Step-by-step procedural summary**

Pilot testing is conducted entirely online via the Pavlovía platform. Participants complete all procedures remotely on their own computers. All research is conducted by Nicholas Meister (PI), supervised by Dr. Krystyna Matusiak, with no in-person contact between researchers and participants. Full step-by-step detail is provided in Section B.4 above. Steps in brief:

1. Eligibility screening via SONA (~ 2 min)
2. Informed consent (~ 2 min)
3. Survey 1: Sensibility Judgment Task (~ 10 min) — computerized priming task with F/J keyboard responses; randomized trial order; response time and accuracy recorded
4. Between-survey break and transition screen (~ 2 min)
5. Survey 2: LCSH Appropriateness Rating Task (~ 28 min) — simultaneous sentence pair and heading display; 7-point rating scale; heading order randomized
6. Demographics (~ 1 min)
7. Qualitative feedback survey (~ 4 min) — anonymous; does not affect SONA credit
8. Debriefing

Survey order randomized between participants. Total session duration: approximately 45–55 minutes.

B.6.2. H.2.1 — Anticipated risks and discomforts

Physical harms. No physical harms are anticipated. The study involves no physical procedures, invasive measures, drugs, biological specimens, or bodily contact of any kind. All activities are conducted online and consist entirely of reading text and responding via keyboard or mouse. Participants may experience minor eye strain or postural discomfort consistent with any computer-based task of 45–55 minutes duration, which is not greater than that encountered in ordinary daily computer use.

Psychological harms. Psychological risks are minimal and transitory. Participants in Survey 1 may experience mild cognitive fatigue or momentary frustration from the rapid, repeated sensibility judgment task. Some participants may find certain lexically ambiguous sentences mildly confusing to evaluate. Participants in Survey 2 may experience mild uncertainty when evaluating library subject headings whose terminology is unfamiliar to them. None of the stimuli involve sensitive, distressing, or emotionally provocative content; all sentences are drawn from a published, peer-reviewed psycholinguistic dataset (Trott & Bergen, 2023) consisting entirely of neutral materials. The tasks do not involve topics related to trauma, personal behavior, mental health, or any other domain associated with elevated psychological risk. Any discomfort is expected to be transitory and consistent with the ordinary cognitive effort of completing an online research task. The study involves partial withholding of the specific research hypothesis (participants are not informed in advance that LCSH category boundaries are being examined), which is disclosed in full during debriefing.

Privacy and confidentiality. Participants are recruited through SONA, which assigns anonymous participant IDs. SONA identities are not linked to experimental data at any stage; the Pavlovia data pipeline receives only an anonymized ID, structurally preventing re-identification through the experimental record. Eligibility screening does not collect personally identifiable information, and screening data from ineligible participants are not retained. No personally identifiable information — including names, email addresses, or IP addresses — is collected through Pavlovia. The primary residual confidentiality risk is the possibility of a data breach of SONA administrative records, which are managed by the DU Department of Psychology and are outside the PI's direct control. This risk is not greater than that associated with routine SONA use for course credit tracking and is substantially mitigated by the structural separation of SONA identity from experimental data.

Social and/or economic harms. No social or economic harms are anticipated. The study does not collect information about illegal activities, substance use, mental illness, sexual behavior, financial standing, employment, or any other domain that could expose participants to embarrassment, stigmatization, or legal or professional risk. Content is entirely neutral. Participation is voluntary and anonymous; withdrawal carries no academic or professional consequence.

B.6.3. H.2.3 — Risk minimization

Risks are minimized through the following measures:

1. All experimental stimuli are drawn from Trott and Bergen’s (2023) published, peer-reviewed psycholinguistic dataset and consist entirely of short, grammatically neutral sentences with no sensitive or emotionally provocative content.
2. Participation is entirely voluntary and self-paced. Participants may withdraw at any time without penalty to their SONA credit. No follow-up contact is initiated by the PI following study completion.
3. The SONA → Pavlovia pipeline assigns randomized anonymous participant IDs, and no key linking SONA identities to experimental records is created or retained at any point. This structural separation is an inherent feature of the data architecture.
4. Experimental data are stored on Pavlovia’s password-protected servers and, upon download by the PI, are stored on a password-protected, security-software-protected device accessible only to the PI and Faculty Advisor.
5. The total pilot session duration of approximately 45–55 minutes limits exposure to task-related fatigue.
6. The pilot study design itself — collecting diagnostic data from 20 participants before any main data collection — functions as a risk-minimization measure for the subsequent study by enabling identification and correction of technical, procedural, or stimulus-calibration problems before a larger participant pool is exposed to potentially flawed protocols.
7. Expert validation by two DU librarians prior to pilot testing ensures that LCSH stimulus mappings meet professional cataloging standards, reducing the likelihood that participants encounter poorly calibrated or semantically incoherent stimuli.

B.6.4. H.3.1 — Direct benefits

There are no direct therapeutic or diagnostic benefits to individual participants. This is a non-therapeutic behavioral study; participation does not provide medical treatment, psychological intervention, clinical assessment, or individualized feedback of any kind.

Participants may derive modest indirect benefits. Contributing to research that examines whether standard library classification tools align with how users cognitively process language has practical relevance to the usability of information retrieval systems that participants themselves may use as library patrons or students. Participants may also find intellectual interest in the tasks, which engage common language processing abilities in an atypical context. Participants who are undergraduate students in the DU Psychology SONA pool receive research participation credit applicable to course requirements.

At the broader level, the study contributes to foundational knowledge at the intersection of knowledge organization, psycholinguistics, and information retrieval, with potential implications for the design and evaluation of controlled vocabulary systems in library and digital

information contexts.

Part C: Justification for Exempt Certification

Scope of this section. This section provides the full justification for exempt certification of the capstone study under 45 CFR 46.104(d) (2018 Requirements). It covers all three study components: proof-of-concept analysis of publicly available existing data (Component A), expert consultant validation of LCSH stimulus mappings by DU librarians (Component B), and pilot testing of two online behavioral surveys with SONA participants (Component C). The PI does not in any way claim authority to make an independent exemption determination; this justification is submitted for certification by the DU IRB under Protocol #2446127-1.

C.1. Threshold Requirements: Minimal Risk and IRB Submission Requirement

The DU IRB requires that each study satisfy the minimal risk threshold (45 CFR 46.102(j)) to be considered for exemption. All three components of this study involve no physical procedures, no sensitive personal disclosures, no clinical assessments, and no stimuli that could plausibly cause psychological distress. The maximum discomfort anticipated is mild cognitive fatigue from reading sentences and pressing keyboard keys, which is ordinary to daily computer use. The minimal risk threshold is satisfied for all components. This application is submitted as a request for exempt certification.

C.2. Component A: Proof-of-Concept Analysis of Publicly Available Existing Data

The first study component involves analysis of Trott and Bergen’s (2023) experimental data, which is publicly available on GitHub. DU IRB’s Levels of Review guidance confirms that Category 4 exemption applies when “the information/biospecimens are publicly available,” directly tracking 45 CFR 46.104(d)(4)(i). The Trott and Bergen dataset fully satisfies this condition: it is deposited in a public repository, openly accessible without restriction, and was collected under a previously approved IRB protocol at the original investigators’ institution (UC San Diego). No new interaction with human subjects occurs during this component; the PI accesses and analyzes a pre-existing, de-identified, publicly available dataset. No consent from the original participants is required or obtainable, and none is sought.

The purpose of this component is to validate key independent variables (`bert_distance`, `ambiguity_type`) and the cross-classified mixed-effects model specification, confirming that the analysis pipeline behaves as expected before any new data collection begins.

C.3. Component B: Expert Consultant Validation of LCSH Mappings by DU Librarians

As established in Part A of this document, because the librarians are engaged solely in a professional expert-consultation capacity and are not the object of investigation, this activity is not considered human-subjects research under 45 CFR 46.102(e)(1). No private information, behavioral data, or personal characteristics are collected from them. Confidentiality protections described in Part A of this document are maintained as a matter of professional ethics and institutional best practice.

C.4. Component C: Pilot Testing of Online Surveys with SONA Participants

The pilot testing component involves human subjects and is the primary focus of this exemption justification. It comprises two computerized tasks — a sensibility judgment task (Survey 1) and an LCSH appropriateness rating task (Survey 2) — followed by an anonymous qualitative feedback survey. These activities qualify for exempt certification under **two independent and overlapping exemption categories**: 45 CFR 46.104(d)(2) and 45 CFR 46.104(d)(3). Either category is independently sufficient; both are argued in the alternative.

C.4.1. Exemption Category 2: Survey and cognitive test procedures

Under 45 CFR 46.104(d)(2), research is exempt if it “only includes interactions involving educational tests (cognitive, diagnostic, aptitude, achievement), survey procedures, interview procedures, or observation of public behavior (including visual or auditory recording)” and at least one of three criteria is met:

- (i) The information obtained is recorded by the investigator in such a manner that the identity of the human subjects cannot readily be ascertained, directly or through identifiers linked to the subjects;
- (ii) Any disclosure of the human subjects’ responses outside the research would not reasonably place the subjects at risk of criminal or civil liability or be damaging to the subjects’ financial standing, employability, educational advancement, or reputation; or
- (iii) The information obtained is recorded in such a manner that identity can be ascertained and an IRB conducts a limited IRB review to make the determination required by §46.111(a)(7).

Applicability to Survey 1 (Sensibility Judgment Task). Survey 1 presents sentence pairs and collects response-time and accuracy data via keyboard responses. This is a standard paradigm in cognitive psycholinguistics and constitutes a *cognitive test* under 45 CFR

46.104(d)(2). The regulation’s reference to “cognitive” tests is broad by design and encompasses behavioral tasks measuring language processing and semantic interpretation. DU IRB’s Levels of Review guidance confirms that Expedited Review Category 7 covers “research on individual or group characteristics or behavior (including, but not limited to, research on perception, cognition, motivation, identity, language, communication, cultural beliefs or practices, and social behavior),” demonstrating that cognitive and language tasks of this type are squarely within the behavioral research the IRB classifies at the minimal-risk end of the spectrum. The sensibility judgment task is, if anything, less intrusive than the paradigms associated with expedited review, warranting exempt rather than expedited status.

Applicability to Survey 2 (LCSH Appropriateness Rating Task). Survey 2 presents sentence pairs with proposed subject headings and collects 7-point Likert-scale ratings. This is straightforwardly a **survey procedure** — participants read stimuli and record scalar responses to each item, which is the definitional form of survey research. The DU IRB Category 2 description explicitly encompasses online surveys.

Applicability to the qualitative feedback survey. The post-pilot-test qualitative feedback survey is an anonymous online survey procedure and satisfies Category 2 independently and as part of the overall study package.

Anonymity criterion: 45 CFR 46.104(d)(2)(i). All three survey and task components satisfy the first anonymity criterion. Participants are recruited via SONA, which assigns anonymous IDs. DU IRB defines anonymous data as data that “can never be linked to the subject (directly or indirectly) either because the name/identity of the subject is never obtained by the investigator, or because there is no code linking the data to the subject’s name/identity.” The SONA → Pavlovia pipeline in this study satisfies this definition: the Pavlovia data pipeline receives only an anonymized ID, and no key linking SONA identity to experimental records is created or retained at any point. The PI cannot link any experimental record to a named individual. The information is therefore recorded in such a manner that the identity of human subjects cannot readily be ascertained, directly or through identifiers linked to the subjects.

Alternative criterion. Even if the anonymity criterion were not fully satisfied, 45 CFR 46.104(d)(2)(ii) provides an independent basis. The subject matter of this study — judging sentence sensibility and rating the appropriateness of library subject headings — cannot plausibly expose participants to criminal or civil liability, or damage their financial standing, employability, educational advancement, or reputation. The content is entirely neutral, drawn from a published psycholinguistic dataset. Criterion (ii) is satisfied independently.

C.4.2. Exemption Category 3: Benign behavioral interventions

As an independent alternative basis for exemption, both surveys also qualify under 45 CFR 46.104(d)(3), which was added to the Common Rule in the 2018 revision specifically to ac-

commodate minimal-risk behavioral paradigms that involve an intervention element beyond mere survey or test administration.

Under 45 CFR 46.104(d)(3)(i), research is exempt when it involves “benign behavioral interventions in conjunction with the collection of information from an adult subject through verbal or written responses (including data entry) or audiovisual recording if the subject prospectively agrees to the intervention and information collection” and at least one of the same anonymity or risk criteria as Category 2 is met.

Definition of benign behavioral intervention. The regulation at 45 CFR 46.104(d)(3)(ii) provides:

For the purpose of this provision, benign behavioral interventions are brief in duration, harmless, painless, not physically invasive, not likely to have a significant adverse lasting impact on the subjects, and the investigator has no reason to think the subjects will find the interventions offensive or embarrassing. Provided all such criteria are met, examples of such benign behavioral interventions would include having the subjects play an online game, having them solve puzzles under various noise conditions, or having them decide how to allocate a nominal amount of received cash between themselves and someone else.

Survey 1 maps directly onto the “solving puzzles under various noise conditions” example: participants read sentence pairs and make rapid sensibility judgments under the experimental manipulation of sense-boundary conditions. The task is brief (approximately 8–10 minutes), harmless, painless, not physically invasive, and produces no lasting adverse effects. No participant could reasonably find judging whether a sentence “makes sense” to be offensive or embarrassing. SACHRP guidance further clarifies that “the brief in duration requirement only pertains to the intervention itself, and not the data collection,” meaning the 8- to 10-minute session length does not disqualify the task provided individual trial-level interventions are brief (each trial runs approximately 3–6 seconds).

Survey 2 similarly qualifies: rating the appropriateness of subject headings against sentence contexts is a brief, harmless, non-invasive written-response task presenting no conceivable physical, psychological, or reputational risk.

Prospective agreement requirement. Both surveys require participants to prospectively agree to the intervention and information collection through the informed consent process before any task begins, satisfying the prospective agreement requirement of 45 CFR 46.104(d)(3)(i).

Anonymity criterion: 45 CFR 46.104(d)(3)(A). As described for Category 2 in Section C.4.1, the SONA → Pavlovia pipeline produces data that cannot readily be linked to subject identity, satisfying 45 CFR 46.104(d)(3)(A) in parallel with 45 CFR 46.104(d)(2)(i).

Deception provision: 45 CFR 46.104(d)(3)(iii). The regulation specifies that “if the research involves deceiving the subjects regarding the nature or purposes of the research, this exemption is not applicable unless the subject authorizes the deception through a prospective

agreement to participate in research in circumstances in which the subject is informed that he or she will not be told the specific study hypothesis until after participation.”

This study involves partial withholding of the specific research hypothesis — participants are not told in advance that LCSH categorical boundaries are predicted to correlate with response-time patterns, to prevent demand effects — but they are accurately informed that the study examines how people understand words with multiple meanings and involves evaluating library subject headings. The nature and purposes of the research are accurately disclosed; only the specific directional hypothesis is withheld. This is structurally distinguishable from “deception regarding the nature or purposes of the research” in the regulatory sense.

Nevertheless, to ensure this provision is unambiguously satisfied, the consent form includes language notifying participants that some aspects of the study’s specific hypotheses will be explained fully in the debriefing at the end of the session, and participants prospectively agree to this arrangement before proceeding. This satisfies 45 CFR 46.104(d)(3)(iii) on its own terms.

C.5. Procedural Compliance: IRB Submission and Review Requirements

DU IRB FAQ confirms: “Only the IRB may make the determination that research is exempt from IRB review. Therefore, the investigator must submit the Request for Exemption application to the IRB, which details the research.” This study was originally submitted to the DU IRB under Protocol #2415313-1 in February 2026 and was subsequently withdrawn. It is superseded by the current submission (including a much reduced scope) which constitutes the application for which exempt certification is sought. The PI has not begun any component of the study involving human subjects prior to obtaining certification.

DU IRB further confirms that for non-federally funded projects classified as exempt, “a two-year time period is applied to the study,” and these studies “are not issued expiration dates.” This capstone study is non-federally funded; no annual continuing review will be required following issuance of exempt certification, though the PI will notify the IRB upon completion of all study activities per standard DU practice.

C.6. Summary: Exemption Classification by Study Component

Key principle. The OHRP Decision Charts for the 2018 Requirements confirm: “To be exempt, no nonexempt activities can be involved. Research that includes both exempt and nonexempt activities is not exempt. Research may involve activities exempt under more than one exemption category.” Each component of this study is classified below; no component is nonexempt.

Component	Human sub-jects?	Governing exemption	Criterion satisfied
Analysis of Trott & Bergen (2023) OSF dataset	Yes (existing)	45 CFR 46.104(d)(4): publicly available secondary research	No consent required; dataset publicly available without restriction
Librarian consultant validation (2 DU librarians)	No — professional capacity	45 CFR 46.102(e)(1)	See Part A: librarians are engaged solely in a professional expert-consultation capacity and are not the object of investigation
Survey 1: Sensibility Judgment Task (SONA participants)	Yes	Primary: 46.104(d)(2) cognitive test. Alternative: 46.104(d)(3) benign behavioral intervention	46.104(d)(2)(i) and (d)(3)(A): data anonymous by structural SONA/Pavlovia separation. Also 46.104(d)(2)(ii): no legal, reputational, or financial risk
Survey 2: LCSH Appropriateness Rating Task (SONA participants)	Yes	Primary: 46.104(d)(2) survey procedure. Alternative: 46.104(d)(3) benign behavioral intervention	Same as Survey 1
Qualitative feedback survey (SONA participants)	Yes	45 CFR 46.104(d)(2) survey procedure	46.104(d)(2)(i): anonymous by design; cannot be linked to subject identity

Part D: Justification for Use of PsychoPy and Pavlovia

The principal investigator respectfully submits the following justification for the use of PsychoPy (behavioral experiment software) and Pavlovia (online hosting platform) in the pilot testing phase of this study, and in the full experiments to be submitted under a subsequent IRB application. The IRB’s comment noted that DU holds vetted subscriptions to Qualtrics and REDCap and encouraged their use. The principal investigator appreciates that guidance and addresses it directly below. The short answer is that Qualtrics and REDCap are survey and clinical data capture platforms; Survey 1 of this study is a response-time task. These are categorically different tools for categorically different scientific purposes, and substituting one for the other would invalidate the study’s primary dependent variable. PsychoPy and Pavlovia are the established methodological standards for this class of research and are in active use by the University of Denver Psychology Department.

D.1. Technical Necessity: Qualtrics and REDCap Cannot Implement the Survey 1 Paradigm

Survey 1 of this study is not a survey in the sense for which Qualtrics is designed. It is a *primed sensibility judgment task* — a behavioral experiment in which the dependent variable is participants’ response time (RT), measured in milliseconds from stimulus onset to keypress. Participants read sentence pairs and press a key to indicate whether a sentence is sensible; the latency of that keypress is the primary measurement. This paradigm is a direct methodological replication of Trott and Bergen (2023, *Psychological Review*) and is standard in experimental psycholinguistics.

The task requires:

- (a) precise control of stimulus timing and display duration;
- (b) millisecond-accurate RT recording referenced to stimulus onset;
- (c) keyboard response capture independent of browser rendering delays; and
- (d) randomized trial presentation with within-experiment counterbalancing.

Qualtrics is a survey platform designed for self-report data collection. It does not support stimulus timing control, does not provide millisecond-precision RT measurement, and does not support keyboard-response paradigms of this type. Faculty in the DU Psychology Department who have attempted to use Qualtrics for cognitive tasks report that “the timing resolution is not good and extracting cognitive performance from the output data is a real pain” (personal communication, 2026). REDCap is a clinical data capture platform opti-

mized for structured forms, longitudinal data, and medical research workflows; it similarly lacks the timing infrastructure required for behavioral experiments. Using either platform for Survey 1 would render the response time data scientifically invalid and would make replication of Trott and Bergen's (2023) validated methodology impossible.

PsychoPy (an open-source package for running experiments in Python) is the established methodological standard for this class of experiment. It was developed specifically for behavioral and cognitive research, provides frame-accurate stimulus timing, and records RT referenced to stimulus onset with sub-millisecond precision. Pavlovia is PsychoPy's companion online hosting platform, which allows PsychoPy experiments to run in participants' browsers while preserving the timing precision required for RT research. The combination of PsychoPy and Pavlovia is the accepted infrastructure for online behavioral experiments in cognitive psychology and psycholinguistics.

Survey 2 (LCSH appropriateness ratings on a 1–7 scale) does not require RT measurement and could in principle be implemented in Qualtrics. However, implementing Survey 2 in a separate platform would create an inconsistent participant experience, require participants to navigate two different systems in a single session, and complicate data management and linkage between the two surveys. For methodological coherence, both surveys are implemented within the same PsychoPy/Pavlovia environment.

Summary of platform roles. Qualtrics and REDCap are appropriate for surveys, structured forms, and clinical data capture. PsychoPy and Pavlovia are the standard tools for precisely-timed behavioral tasks. The IRB's question implicitly treats these as interchangeable; they are not. The principal investigator respectfully requests that the IRB evaluate the choice of PsychoPy and Pavlovia against any specific human subjects concerns that arise from their use, rather than against whether a DU-licensed survey platform could nominally collect some form of data.

D.2. Existing Use at the University of Denver

PsychoPy and Pavlovia are in active use by the University of Denver Psychology Department for online behavioral research. Multiple faculty members and their graduate students are currently running experiments with these platforms, and the Department used PsychoPy and Pavlovia to administer online cognitive tasks during the COVID-19 period when all in-person research was suspended. As one faculty member stated, these are “standard tools for our science” and “other labs at DU use these standard tools” (personal communication, 2026).

The principal investigator consulted two DU Psychology Department faculty members in preparing this response. Both confirmed:

1. PsychoPy is in active use in the Psychology Department for both in-lab and online experiments;

2. Pavlovia is the appropriate companion hosting platform for online PsychoPy experiments;
3. Qualtrics and REDCap are not suitable substitutes for precisely-timed cognitive tasks; and
4. PsychoPy/Pavlovia is the standard methodological infrastructure for this class of research.

The Psychology Department’s established use of these platforms at DU constitutes clear institutional precedent. The principal investigator is prepared to work with the Technology Solution Center to complete any formal technology review process required and will provide the IRB with documentation of that review upon its completion. If the Technology Solution Center confirms that PsychoPy and Pavlovia are already in approved use at the institutional level through the Psychology Department, the principal investigator requests that this approval be recognized as applicable to the present study.

D.3. Data Security and Privacy Practices

Pavlovia is operated by Open Science Tools Ltd., a not-for-profit organization based in the United Kingdom. Data collected through Pavlovia are stored on servers compliant with the General Data Protection Regulation (GDPR), which provides data protection standards equivalent to or exceeding those applicable under U.S. research regulations for the minimal-risk data collected in this study.

Data collected in this study are limited to anonymous reaction time measurements, anonymous keypress responses, and anonymous demographic survey responses (age, gender, native language, and mobile device use). No names, email addresses, student ID numbers, IP addresses, or other directly identifying information are collected through Pavlovia. Participants are assigned anonymous subject codes at the point of consent. All data exported from Pavlovia are stored in a password-protected folder accessible only to the principal investigator.

One narrow exception concerns participant identification for research credit purposes. Participants are recruited through SONA Systems, through which undergraduate students may earn course credit for participation. To facilitate credit assignment, SONA Systems captures participants’ identifying information (name and student ID) at the point of sign-up. This information is handled entirely within SONA Systems. Critically, it is *architecturally separated* from experimental data: it is never linked to individual response records or reaction time measurements, and it is not accessible to the principal investigator. The principal investigator receives only anonymous experimental data files from Pavlovia. The separation between participation records (SONA Systems) and experimental data (Pavlovia) is a structural feature of the SONA–Pavlovia integration, not a procedural commitment that depends on researcher behavior. This design ensures that the principal investigator cannot, even in

principle, identify which participant produced which set of responses.

The data collected in this study present no greater privacy risk on Pavlovia than they would on Qualtrics or REDCap. None of the platforms collect sensitive personal information, health information, or information that could expose participants to harm if disclosed.

D.4. Faculty Contact Information

The following DU Psychology Department faculty member (who is not connected to this study) has confirmed the active departmental use of PsychoPy and Pavlovia and is willing to speak with the IRB directly if requested.

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D.5. Summary

PsychoPy and Pavlovia are required for this study because Survey 1 depends on millisecond-precision RT measurement that Qualtrics and REDCap cannot provide. These platforms are the established standard for online behavioral experiments in cognitive psychology, are in active use by the DU Psychology Department, and collect only anonymous, minimal-risk data under GDPR-compliant storage practices. The principal investigator is prepared to complete the DU technology review process if necessary and will provide documentation to the IRB upon its completion.

End of Supplemental Procedures Document (Parts A, B, C, and D). IRBNet Protocol #2446127-1. Nicholas Meister, PI. Krystyna Matusiak, Faculty Sponsor. University of Denver, Morgridge College of Education.